Vince Ong

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Technical Skills

Programming Languages/ libraries: C++, Python, TypeScript, JavaScript, Tailwind, CSS, SQL, MATLAB, React
Software & Tools: Git/GitHub, CI/CD, Docker, LLM's Supabase, Vercel, Antigravity IDE, LTspice, EAGLE
Hardware: PCB Assembly, Microcontrollers, Soldering, Oscilloscopes, Function Generators, DMM, 3D Printing

Experience

Resilient Privacy **July 2026 – Present**
Software Engineer Intern *Dallas, TX (Remote)*

- Audited and optimized **20+** core components within the platform's **UI kit**, resolving visual inconsistencies and streamlining design patterns to accelerate future frontend development cycles by **20%**.
- Engineered scalable backend **APIs** and optimized relational **SQL** database schemas, implementing strategic indexing to ensure robust data delivery and high-performance querying for a React frontend.
- Fortified production infrastructure by authoring rigorous unit and integration tests, and championed professional **Git** workflows to guarantee secure-by-default system deployments.

SFU Racerbot (Autonomous Racing Team) **June 2026 – Present**
Software Developer *Burnaby, B.C.*

- Contributed on a C++/Python sensor fusion pipeline using **ROS2** to combine 2D LIDAR and camera input for obstacle detection.
- Validated autonomous control algorithms in the **F1TENTH Gym simulator**, streamlining physical testing and mitigating safety-critical bugs prior to hardware deployment.

TELUS Digital AI **June 2026 – Present**
Online Data Analyst *Remote*

- Processed and structured large-scale geographical datasets to optimize **training data pipelines** for spatial AI/ML mapping systems.
- Executed data validation protocols across distributed sources to resolve discrepancies and extract edge-case anomalies for downstream **machine learning algorithms**.
- Analyzed and evaluated generative text-to-speech (TTS) models, assessing audio data outputs for phonetic accuracy to improve the performance of **conversational AI systems**.

Projects

Engram ↗ | *TypeScript, Python, Supabase* **April 2026 – Present**
End-to-End Developer | Althra x TechTO Hacks Finalist; Best use of backboard.io

- Deployed a B2B AI SaaS unifying fragmented enterprise data, **reducing employee onboarding time by 30%** through seamless integration of LLM-powered persistent memory.
- Engineered a scalable semantic data pipeline with Supabase and backboard.io, **indexing over 10,000 records** into knowledge graphs to enable voice and text search.
- Architected a custom RAG-driven search interface built in TypeScript, optimizing context-aware retrieval logic across multi-source knowledge bases to consistently deliver **sub-200ms query responses**.

Embedded Circadian Telemetry System ↗ | *C++, I2C, EEPROM* **May 2025 – Present**
Independent Embedded Systems Project

- Engineered a state-driven embedded C++ firmware architecture, utilizing **non-blocking timing arrays** to concurrently process asynchronous sensor polling and hardware interrupts without halting continuous main loop execution.
- Integrated diverse hardware peripherals across a shared I2C bus, developing a **non-linear PWM algorithm** utilizing floating-point cosine math to precisely modulate solid-state transistors and simulate natural sunrise illuminance.

Algorithmic Web Runner ↗ | *HTML5, JavaScript, CSS* **April 2026**
Fullstack Developer | MLH Hack Week Winner

- Built a procedurally generated web application in HTML5 and JavaScript, sustaining a **flawless 60 FPS rendering baseline**.
- Optimized complex asynchronous state management architecture supporting **over 100 concurrent users** with a **P95 latency consistently under 50ms**.
- Integrated an algorithmic audio engine utilizing object-oriented programming principles, modulating digital soundscapes in real-time based on continuous user input events.

Education

Simon Fraser University **September 2025 – Present**
Bachelor of Applied Science in Mechatronic Systems Engineering – Cumulative GPA: 3.7/4.3 *Burnaby, B.C.*

- BC Achievement Scholarship | *Awarded to the top 3% of BC Graduates* *October 2025*
- International Baccalaureate Diploma | *Graduated with an IB 35/45* *July 2025*